

V4.2 Toy Model 5A

Galactic-Scale Relational Tension and Dark Matter Analogue

Rev A - Dark Sector / Rotation-Curve Stress Test

Date: 2026.05.26 | Continuation after Gravity Track 4A-4J Close-Out

Status: Toy Model 5A is a galactic-scale toy model. It tests whether an added relational-tension term can mimic the rotation-curve signature normally attributed to a dark-matter halo. It does not claim to disprove particle dark matter, replace Lambda-CDM, fit real galaxies, explain lensing surveys, explain the CMB, or prove physical 3T.

1. Purpose

Toy Model 4A-4J closed the weak-field gravity baseline. Toy Model 5A opens the Dark Sector track by asking a controlled question: can large-scale relational graph tension produce the same mathematical signature as a dark-matter halo, namely a flattened outer galactic rotation curve?

The test is deliberately simple. It does not use real galaxy data. It compares three curves: a visible-source-only curve, a Keplerian falloff reference, and a visible-source-plus-relational-tension curve.

2. Guardrail

This model does not show that dark matter is false. It shows that a specific relational-tension term can mathematically reproduce the rotation-curve effect that would be interpreted as extra mass under Newtonian analysis. Dark matter has many other observational constraints: gravitational lensing, galaxy clusters, large-scale structure, and the cosmic microwave background. Those are not tested here.

3. Visible-source rotation curve

The visible baryonic source is represented by a smooth disk-like proxy:

$$v_b^2(r) = G M_b r^2 / (r^2 + a_b^2)^{3/2}$$

Meaning of symbols:

- $v_b(r)$ is the circular speed predicted from the visible source only.
- G is the dimensionless gravitational coupling used in the toy model.
- M_b is the total visible/baryonic source scale.
- a_b is the softening/disk scale that prevents a singular center.
- At large r , this visible-only contribution falls approximately like v proportional to $1/\sqrt{r}$, the Keplerian decline.

4. Relational-tension term

For a flat rotation curve, the required outer acceleration behaves like $a(r)$ proportional to $1/r$, because $v^2 = r a$. Toy Model 5A models large-scale relational graph tension as a smooth term that activates away from the galactic center:

$$v_{rel}^2(r) = V_{inf}^2 [1 - \exp(-r/R_t)]$$

Meaning of symbols:

- $v_{rel}(r)$ is the circular-speed contribution from relational tension.
- V_{inf} is the asymptotic flat-speed contribution.
- R_t is the radius over which the relational network tension becomes effective.
- The exponential activation keeps the center regular and prevents the relational term from dominating the inner disk.

This is a constitutive dark-sector ansatz, not a derived 3T law. The point of 5A is to test the signature, not to claim final physics.

5. Total velocity and apparent halo mass

$$v_{total}^2(r) = v_b^2(r) + v_{rel}^2(r)$$

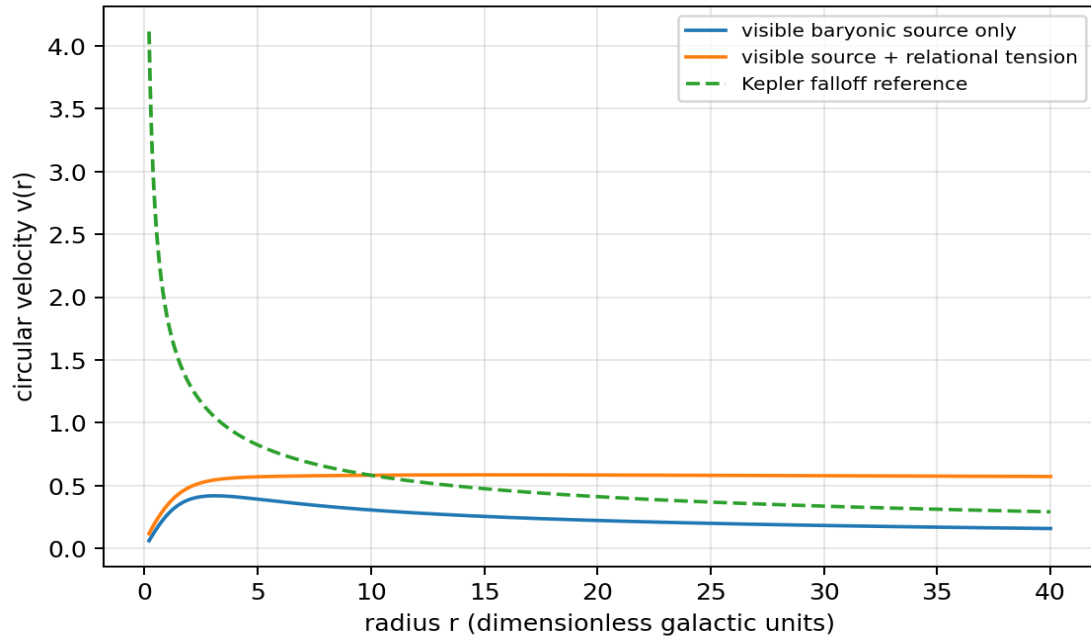
$$M_{eff}(r) = v_{total}^2(r) r / G$$

$$M_{rel}(r) = v_{rel}^2(r) r / G$$

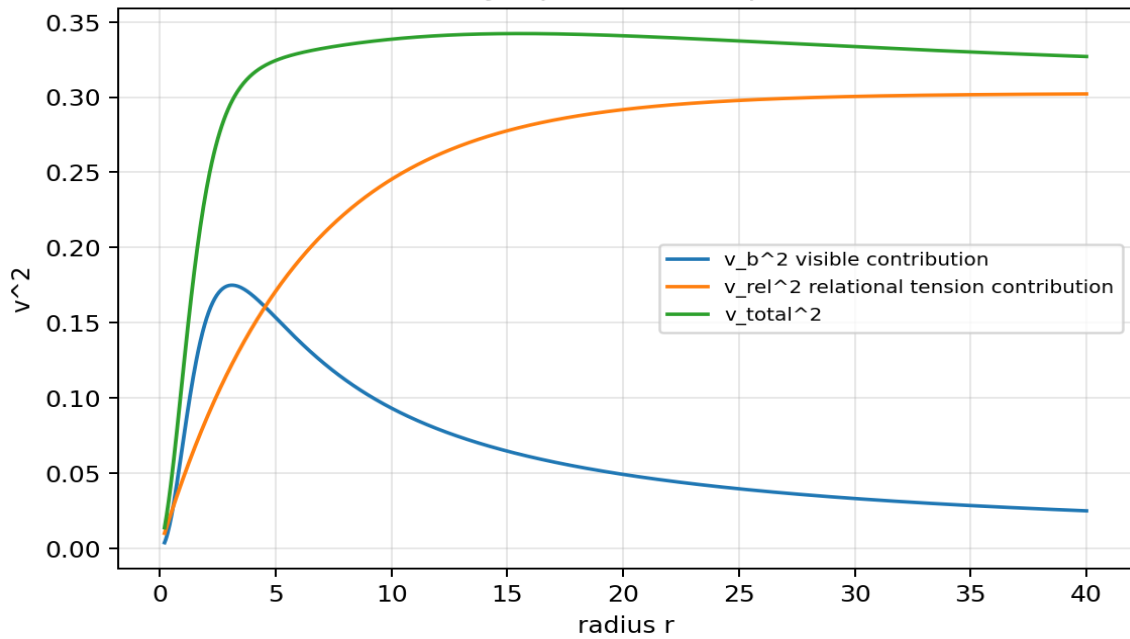
If an observer assumes only ordinary Newtonian mass, the relational-tension contribution would be inferred as extra missing mass. In the outer region, M_{rel} grows approximately linearly with r , which is the classic effective behavior needed for flat

rotation curves.

Toy 5A rotation curve: Kepler decline versus relational-tension flattening



Velocity-squared decomposition



6. Rotation-curve result

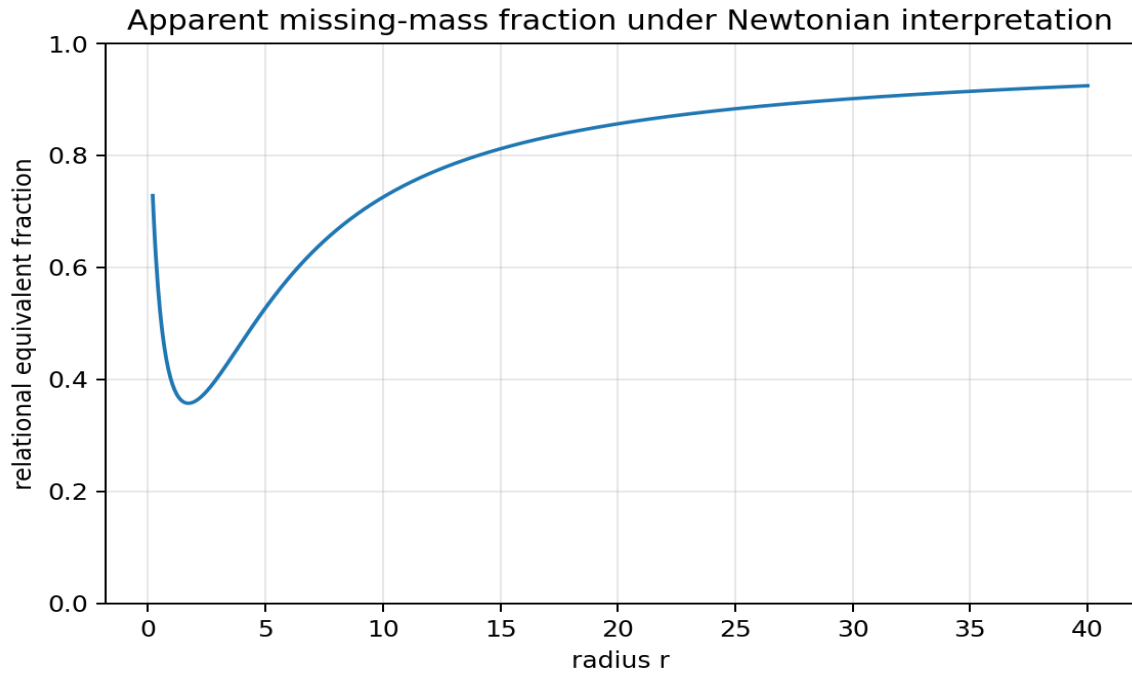
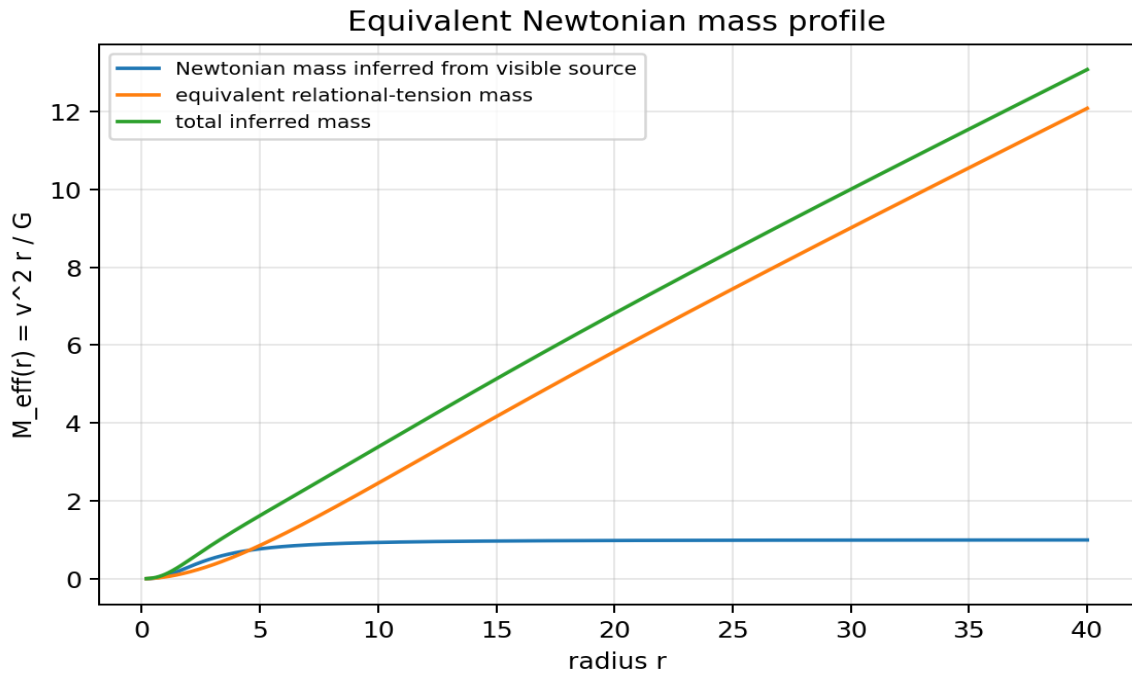
Diagnostic	Value	Interpretation
Outer log slope of visible-only $v(r)$	-0.4909	Close to Keplerian decline.
Outer log slope of relational $v_{\text{rel}}(r)$	0.0224	Approaches flat contribution.
Outer log slope of total $v(r)$	-0.0312	Nearly flat outer rotation curve.
Visible-only outer coefficient of variation	0.1376	Outer visible-only speed keeps changing.
Total outer coefficient of variation	0.0073	Outer combined speed is much flatter.
Relational mass linearity R^2 , $r=15..40$	0.999791	M_{rel} approximately proportional to r .
Equivalent density outer log slope	-2.1043	Near $1/r^2$ halo-like profile.

Result: the visible-only curve declines, while the relational-tension curve stays nearly flat at large radii.

7. Equivalent missing-mass interpretation

A Newtonian observer who does not know about the relational-tension term would infer an extra mass component:

$$M_{\text{rel}}(r) = v_{\text{rel}}^2(r) \, r / G$$



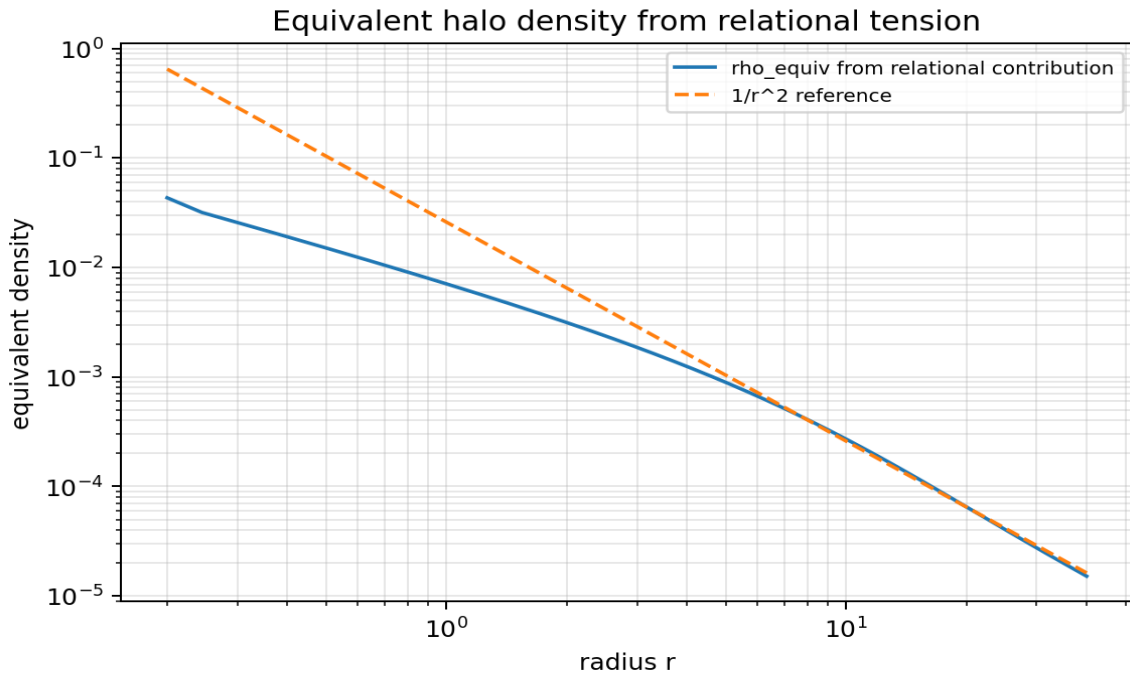
r	v_b	v_{rel}	v_{total}	$M_{\text{visible_eff}}$	$M_{\text{rel_equiv}}$	rel. fraction
5.0	0.3916	0.4136	0.5696	0.7668	0.8552	0.527
10.0	0.3052	0.4953	0.5818	0.9316	2.4537	0.725
15.0	0.2541	0.5269	0.5850	0.9686	4.1650	0.811
20.0	0.2216	0.5401	0.5838	0.9821	5.8342	0.856
30.0	0.1818	0.5481	0.5775	0.9920	9.0139	0.901
40.0	0.1578	0.5496	0.5718	0.9955	12.0846	0.924

8. Equivalent density and acceleration

The equivalent spherical density of the relational contribution is computed from:

$$\rho_{\text{equiv}}(r) = [1 / (4 \pi r^2)] \, dM_{\text{rel}}/dr$$

In the outer region, the equivalent density approaches an approximately $1/r^2$ profile, which is the same broad scaling used by isothermal halo models to produce flat rotation curves.



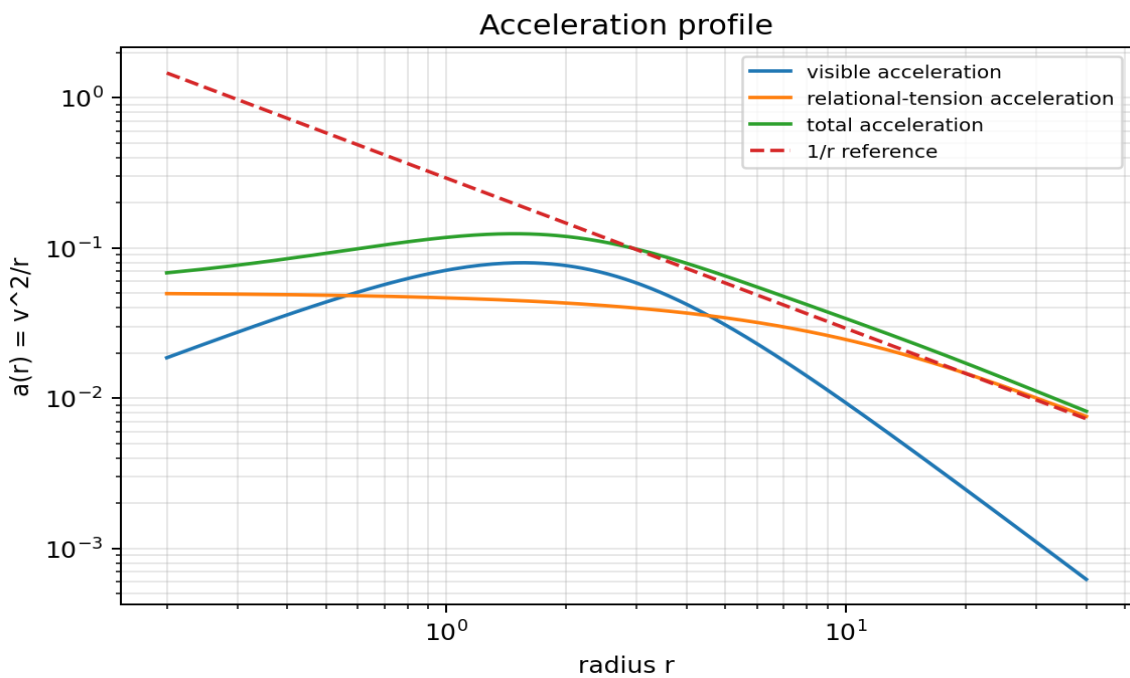
The acceleration view is even simpler:

$$a_{\text{rel}}(r) = v_{\text{rel}}^2(r) / r$$

At large r :

$v_{\text{rel}}^2 \rightarrow \text{constant}$

therefore $a_{\text{rel}} \rightarrow 1/r$



9. Interpretation inside V4.2

Toy Model 5A supports this limited ladder:

- visible galactic source
 - + large-scale relational graph tension
 - > extra inward acceleration approximately proportional to $1/r$
 - > flat outer rotation curve
 - > Newtonian observer infers a halo-like missing mass profile

It does not support this stronger ladder:

- relational graph tension
 - > all dark matter observations solved
 - > Lambda-CDM replaced
 - > particle dark matter disproved
 - > physical 3T proven

10. What Toy Model 5A achieves

- It opens the Dark Sector track with a controlled galactic rotation-curve test.
- It shows that an outer relational-tension term can mathematically flatten a rotation curve.
- It shows the Newtonian equivalent of that term behaves like an extended halo mass.
- It shows the equivalent outer density approaches a $1/r^2$ -like scaling.
- It provides a clean bridge from the 4A-4J weak-field baseline to a dark-matter-style observational signature.

11. What Toy Model 5A does not achieve

- It does not derive the relational-tension term from first principles.
- It does not fit real galaxy data.
- It does not reproduce the baryonic Tully-Fisher relation.
- It does not address galaxy-cluster lensing.
- It does not address the Bullet Cluster.
- It does not address cosmological structure formation.
- It does not address the cosmic microwave background.
- It does not disprove particle dark matter.
- It does not prove physical 3T.

12. Close-out statement for 5A

Toy Model 5A is a successful first dark-sector analogue test. A simple large-scale relational-tension contribution produces the same primary mathematical signature that motivates dark-matter halos in galaxies: a flat outer rotation curve and an inferred mass profile that keeps growing with radius. The result is not a dark-matter theory. It is a compatibility result showing that relational graph tension can mimic the rotation-curve effect in a controlled toy setting.

13. Recommended next chapter

Recommended next: Toy Model 5B - Derive the Relational Tension Term from Graph Edge Economics.

- Do not assume $v_{\text{rel}}^2 = \text{constant}$ at large r .
- Start from a ring graph / radial graph with baseline edge tension.
- Derive whether accumulated edge tension naturally gives $M_{\text{rel}}(r)$ proportional to r .
- Then test whether the baryonic Tully-Fisher scaling can emerge.
- Only after 5B should real galaxy data or lensing analogues be attempted.

14. Rebind prompt for continuation

We are resuming after Toy Model 5A.

5A opened the Dark Sector track after the Gravity Track 4A-4J close-out.

Question:

Can large-scale relational graph tension mimic the galactic rotation-curve signature usually attributed to dark matter?

Visible-source model:

$$v_b^2(r) = G M_b r^2 / (r^2 + a_b^2)^{3/2}$$

Relational-tension ansatz:

$$v_{\text{rel}}^2(r) = v_{\text{inf}}^2 [1 - \exp(-r/R_t)]$$

Total:

$$v_{\text{total}}^2 = v_b^2 + v_{\text{rel}}^2$$

Equivalent inferred mass:

$$M_{\text{eff}} = v_{\text{total}}^2 r / G$$

$$M_{\text{rel}} = v_{\text{rel}}^2 r / G$$

Result:

Visible-only curve declines.

Visible + relational tension curve stays nearly flat.

M_{rel} grows approximately linearly with r .

Equivalent ρ_{rel} approaches roughly $1/r^2$ in the outer region.

Guardrail:

5A does not derive the relational-tension law.

It does not fit real galaxies.

It does not solve lensing, clusters, CMB, structure formation, or dark matter generally.

It does not disprove particle dark matter or prove physical 3T.

Next recommended:

Toy Model 5B - derive the relational-tension term from graph edge economics rather than assuming the flat-velocity contribution.